

Detecting acetonitrile at 90° elevation

A complete example using a LEO downlink

Intuition, observations, matrices and the detection decision

A communication link as a sensing experiment

The transmitter sends a radio signal. The receiver measures how the propagation path has changed it.

This project asks whether those changes also reveal the amount of a gas along the path.

ISAC means integrated sensing and communications: using a communication signal to carry data and obtain information about the environment.

Here the transmitter is on a low Earth orbit (LEO) satellite and the receiver is on the ground.

This is a proposed sensing use of the link. Useful atmospheric detection still needs experimental validation.

The detection question

Has acetonitrile, CH_3CN , increased? Compare a baseline transmission (reference) with a later transmission (sample).

The unknown q is the concentration increase at ground level under an assumed vertical profile, measured in $\mu\text{g}/\text{m}^3$.

LEO links are a useful example throughout the sensing chain. The same reasoning can be adapted to terrestrial and UAV links.

The output is a concentration enhancement and a statistical decision.

One path does not recover an arbitrary vertical profile or a 3D image.

Why a molecule can change a radio signal

Acetonitrile, CH_3CN , is the one gas selected for this example.

Its rotational transitions absorb energy at particular frequencies. Pressure and temperature broaden and modify the spectral features.

A spectral template is the calculated attenuation at every measured frequency for one unit of concentration.

More gas increases the attenuation according to this template, within the assumed profile.

Other molecules can have overlapping signatures. The first example assumes their concentrations do not change.

Many frequencies in one transmitted signal

Term	Physical meaning
OFDM	Orthogonal frequency division multiplexing: many separately recoverable tones
Tone / subcarrier	One regularly spaced frequency component of the waveform
Pilot	A transmitted symbol whose amplitude and phase the receiver already knows
Bandwidth	The total frequency span used at the same time
Symbol	One time interval over which the tone coefficients are transmitted

Center frequency: 235 GHz. Bandwidth: 10 GHz. The example measures 1,024 tones.

How to read the radio power units

Unit	Definition or interpretation	Example
dB	$10 \log_{10}$ of a power ratio	+10 dB means $\times 10$ power
dBm	$10 \log_{10}(P / 1 \text{ mW})$	25 dBm $\approx$ 316 mW
dBi	Antenna gain relative to an isotropic radiator	Directionality, not created energy
Attenuation in dB	Positive loss subtracted in a link budget	10 dB loss leaves 10% of the power

Power ratios multiply in linear units and add in decibels. Amplitude ratios use $20 \log_{10}$.

What the receiver is trying to distinguish

At the first tone, before measurement errors	Model value
Chosen enhancement q for the worked sample	50.5814 $\mu\text{g}/\text{m}^3$
Extra molecular attenuation	0.012181 dB
Sample amplitude / reference amplitude	0.998599
Reduction in amplitude	0.1401%

The measurement must distinguish a small molecular change from noise and instrument drift.

This is the predicted molecular effect for the assumed profile, before adding noise. It is not a measured change.

The intuition

1. Send known tones across a 10 GHz band and measure their received amplitudes.
2. Compare a reference with a sample after correcting known link changes.
3. Match the remaining frequency pattern to the molecular absorption template.
4. Estimate its strength and compare it with a threshold fixed in advance.

The useful evidence is the spectral shape across tones, not one noisy dip.

The 90° geometry

Quantity	Example	Meaning
Ground elevation	90°	Satellite directly overhead
Satellite altitude	550 km	Chosen LEO scenario
Slant range	550 km	Shortest path for this altitude
Atmospheric integration	0–100 km	Standard atmosphere
Concentration profile	$q \exp(-z / 1500 \text{ m})$	Assumed enhancement shape

90° is the maximum ground elevation. A 30° value can be a minimum elevation mask.

This is a static zenith benchmark. A real satellite keeps moving during acquisition.

What the hardware sources demonstrate

Source	Reported quantity	Scope
Sen et al., Nature Electronics	200 mW, about 23 dBm	Measured terrestrial link, 210–240 GHz
TeraLink, 2026 preprint	25 dBm, 7 dB noise figure	Published LEO design, not flight validation
Cooper et al., JMW	>400 mW at 240 GHz	Continuous wave source component

25 dBm is a design anchor for this example, not proof of a complete OFDM payload.

The sources do not demonstrate the full bandwidth, linear output, tracking and calibration together.

The chosen link budget inputs

Input	Value	Status
Total average RF power	25 dBm = 316 mW	Published design anchor
Transmit / receive diameter	0.10 m / 1.0 m	Circular aperture model
Aperture efficiency	0.65 at both ends	Assumed
Gain at 235 GHz	45.96 / 65.96 dBi	Calculated from aperture
Receiver noise figure	7 dB at $T_o = 290$ K	Published design anchor
Implementation loss	5 dB	Assumed budget

The linear average power available for OFDM after amplifier backoff remains unverified.

The OFDM frequency grid

Quantity	Value
Occupied band and tone count K	230–240 GHz, 1,024 tones
Spacing $\Delta f = B / K$	9.765625 MHz
Useful symbol $T_u = 1 / \Delta f$	102.4 ns
Assumed cyclic prefix T_{CP}	10 ns
Transmitted symbol $T_s = T_u + T_{CP}$	112.4 ns
Pilots per 10,000 symbol frame	30 full pilot symbols, 0.3%

All tones share the same 25 dBm total average transmit power.

Twenty seconds include the reference

Acquisition	Scheduled time	Complete frames	Pilots per tone
Reference	10 s	8,896	266,880
Sample	10 s	8,896	266,880

$$M = \left\lfloor \frac{10 \text{ s}}{10,000 \times 112.4 \text{ ns}} \right\rfloor \times 30 = 266,880$$

M is the number of known pilot observations per tone, per acquisition.

19.998208 s are occupied by complete frames. Both acquisitions contribute noise.

Waiting for comparable geometry, calibration and reference availability adds operational time.

The molecular template comes from spectroscopy

$$c(z) = qe^{-z/H_g}, \quad H_g = 1500 \text{ m} \quad a_k = \frac{10}{\ln 10} \int_0^{100 \text{ km}} \sigma_k(z) n_1(z) dz$$

q : surface equivalent enhancement ($\mu\text{g}/\text{m}^3$) · σ_k : effective absorption area per molecule

n_1 : molecular density for $q = 1 \mu\text{g}/\text{m}^3$ · a_k : dB per ($\mu\text{g}/\text{m}^3$). Use consistent length units in the integral.

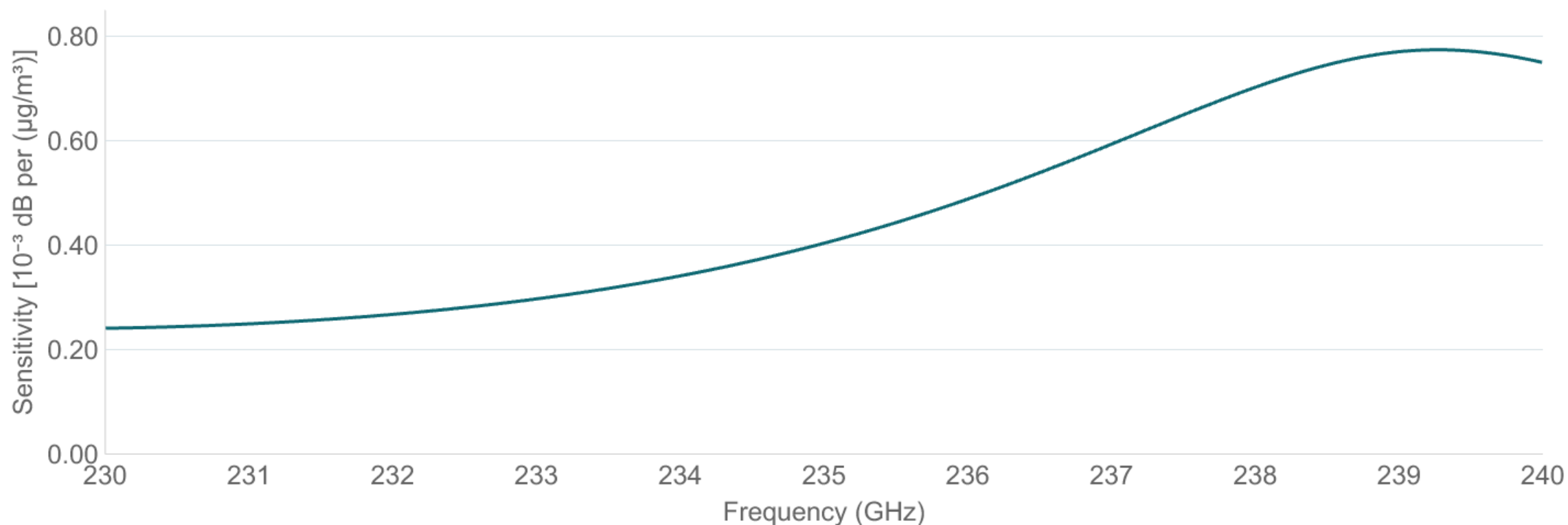
HITRAN supplies CH_3CN line positions, strengths and broadening parameters.

A Voigt model and the atmospheric pressure and temperature give the cross section at each altitude.

The predicted extra attenuation at tone k is $a_k q$ dB.

The vertical shape is assumed. The model does not measure concentration at each height.

The acetonitrile spectral pattern



The estimator uses the full 1,024 tone template, including its curvature.

Calculated absorption from external line parameters, not a measured satellite spectrum.

One tone: the complete power accounting

Term at 230.004883 GHz	Value
Transmit power per tone: $25 - 10 \log_{10}(1024)$	-5.1030 dBm
Transmit gain + receive gain	45.7705 + 65.7705 dBi
Free space loss	194.4898 dB
Oxygen / water attenuation	4.5477 dB
Implementation loss	5.0000 dB
Received reference power	-97.5996 dBm

Received power = tone power + gains – propagation loss – implementation loss.

Receiver noise determines the pilot uncertainty

First tone	Calculated value
Sky temperature	177.831 K
Noise power per tone	-97.4270 dBm
Reference SNR ρ_0 in dB	-0.1726 dB

$$T_e = T_0 \left(10^{NF/10} - 1 \right), \quad N_k = k_B (T_{\text{sky},k} + T_e) \Delta f, \quad \rho_{r,k} = P_{r,k} / N_k$$

NF: receiver added noise, expressed as a noise figure (7 dB) · $T_0 = 290$ K · T_e : equivalent noise temperature
 T_{sky} : sky brightness temperature (K) · k_B : Boltzmann constant · N_k : noise power (W) · ρ : signal to noise ratio

Sky emission and receiver noise are counted once. Noise figure is not applied twice.

The observation matrices

Object	Dimensions	Meaning
$\mathbf{X}_r$	$1,024 \times 266,880$	Known unit magnitude pilots
$\mathbf{h}_r$	$1,024 \times 1$	Amplitude and phase response of each tone
$\text{diag}(\mathbf{h}_r)$	$1,024 \times 1,024$	One complex gain per tone
$\mathbf{Y}_r$ and $\mathbf{W}_r$	$1,024 \times 266,880$	Received samples and complex thermal noise

$$\mathbf{Y}_r = \text{diag}(\mathbf{h}_r) \mathbf{X}_r + \mathbf{W}_r, \quad r \in \{0, 1\}$$

$r = 0$: reference · $r = 1$: sample · k : tone · m : pilot · $|\mathbf{h}_{r,k}|^2$ = received power with unit pilots

Delay, phase and carrier frequency corrections are ideal in this observation model.

Averaging produces one channel estimate per tone

$$\hat{h}_{r,k} = \frac{1}{M} \sum_{m=1}^M Y_{r,k,m} X_{r,k,m}^*, \quad \hat{h}_{r,k} \sim \mathcal{CN}\left(h_{r,k}, \frac{N_k}{M}\right)$$

M = 266,880 pilots per tone in each acquisition · *: complex conjugate
CN: circular complex Gaussian distribution, with stated complex variance

The receiver removes the known pilot phase, then averages coherently.

The raw matrices reduce to two complex vectors, each with 1,024 entries.

Thermal variance falls by M only when the phase correction permits coherent averaging.

Persistent spectral calibration error survives this averaging.

The reference ratio becomes an attenuation vector

$$y_k = -20 \log_{10} \left(\frac{|\hat{h}_{1,k}|}{|\hat{h}_{0,k}|} \right), \quad \mathbf{y} \in \mathbb{R}^{1024}$$

y_k : differential attenuation (dB) · $\hat{h}_0$: reference channel estimate · $\hat{h}_1$: sample channel estimate

Extra molecular absorption reduces the sample amplitude, so the expected y_k is positive.

Known geometry and hardware corrections precede the comparison.

Under matched background, the molecular contribution is $y_k \approx a_k q$.

An amplitude equalizer that removes the molecular pattern would also remove the sensing evidence.

The design matrix separates gas from gain changes

$$\mathbf{y} = \mathbf{A}\boldsymbol{\theta} + \boldsymbol{\varepsilon}, \quad \mathbf{A} = \begin{bmatrix} \mathbf{a} & \mathbf{1} & \mathbf{u} \end{bmatrix} \in \mathbb{R}^{1024 \times 3}$$
$$\boldsymbol{\theta} = \begin{bmatrix} q & b_0 & b_1 \end{bmatrix}^T, \quad u_k = \frac{f_k - \bar{f}}{f_{\max} - f_{\min}}$$

$\mathbf{a}$: molecular template [dB per ($\mu\text{g}/\text{m}^3$)] · q : enhancement ($\mu\text{g}/\text{m}^3$) · $\boldsymbol{\varepsilon}$: remaining measurement error (dB)
 b_0 : gain offset (dB) · b_1 : gain slope coefficient (dB) · $\mathbf{u}$: dimensionless, spanning -0.5 to +0.5

Column 1 is the predicted molecular shape.

Columns 2 and 3 allow an unknown common gain offset and linear spectral tilt.

$\mathbf{A}$ is built from known frequency samples. Only q , b_0 and b_1 are unknown.

The covariance includes both measurements

$$\mathbf{C}(q) = \text{diag}(v_k(q)) + \sigma_{\text{cal}}^2 \mathbf{R}, \quad R_{ij} = e^{-|f_i - f_j|/(10 \text{ GHz})}$$

$$v_k(q) = \frac{(20/\ln 10)^2}{2M} [\rho_{0,k}^{-1} + \rho_{1,k}(q)^{-1}]$$

C: $1,024 \times 1,024$ covariance in dB^2 · v_k : thermal variance of the log ratio

ρ_0 , ρ_1 : linear per pilot SNR · $\sigma_{\text{cal}} = 0.001 \text{ dB}$: assumed differential residual standard deviation

Diagonal thermal noise contains reference and sample uncertainty.

Calibration residual is the instrument error left after correction. Its spectral pattern persists across the pair.

The 0.001 dB residual and its frequency correlation are assumptions, not receiver measurements.

Whitening and removal of nuisance directions

$$\mathbf{C}(0) = \mathbf{L}\mathbf{L}^T, \quad \tilde{\mathbf{y}} = \mathbf{L}^{-1}\mathbf{y}, \quad \tilde{\mathbf{a}} = \mathbf{L}^{-1}\mathbf{a}, \quad \tilde{\mathbf{B}} = \mathbf{L}^{-1}\mathbf{B}$$
$$\mathbf{r} = (\mathbf{I} - \mathbf{Q}\mathbf{Q}^T)\tilde{\mathbf{a}}$$

L: Cholesky factor of null ($q = 0$) covariance · $\mathbf{B} = [1, \mathbf{u}]$ · Q: orthonormal basis for columns of $\mathbf{L}^{-1}\mathbf{B}$

r: the whitened molecular pattern after removing gain offset and slope

Whitening changes coordinates so that the null noise covariance is the identity.

Projection keeps only the molecular pattern that gain offset and slope cannot explain.

If $\mathbf{r}$ were zero, the concentration would be unidentifiable under this nuisance model.

One row of weights estimates concentration

Quantity	Baseline value
Null standard deviation s_0	12.7313 $\mu\text{g}/\text{m}^3$
Response to gas: H a	1, within numerical precision
Response to gain terms: H B	0, within numerical precision

$$\hat{q} = \frac{\mathbf{r}^T \tilde{\mathbf{y}}}{\mathbf{r}^T \mathbf{r}} = \mathbf{H} \mathbf{y}, \quad \mathbf{H} = \frac{\mathbf{r}^T \mathbf{L}^{-1}}{\mathbf{r}^T \mathbf{r}}, \quad s_0^2 = \mathbf{H} \mathbf{C}(0) \mathbf{H}^T$$

H: $1 \times 1,024$ concentration weights $[(\mu\text{g}/\text{m}^3)/\text{dB}]$ · $\hat{q}$: estimated enhancement ($\mu\text{g}/\text{m}^3$)

s_0 : null standard deviation of $\hat{q}$ ($\mu\text{g}/\text{m}^3$)

20 s total • ideal tracking • assumed, unmeasured 0.001 dB differential calibration residual

A fixed threshold turns the estimate into a decision

$$H_0 : q = 0, \quad H_1 : q > 0, \quad \hat{q} > \underbrace{z_{0.99} s_0}_{\tau} = 2.32635 \times 12.7313 = 29.6174 \mu\text{g}/\text{m}^3$$

τ : decision threshold · $z_{0.99}$: standard normal 99th percentile · false alarm target $\alpha = 1\%$

$s_1(q) = \sqrt{[H \ C(q) \ H^T]}$: standard deviation when the enhancement is q , using the fixed null weights H

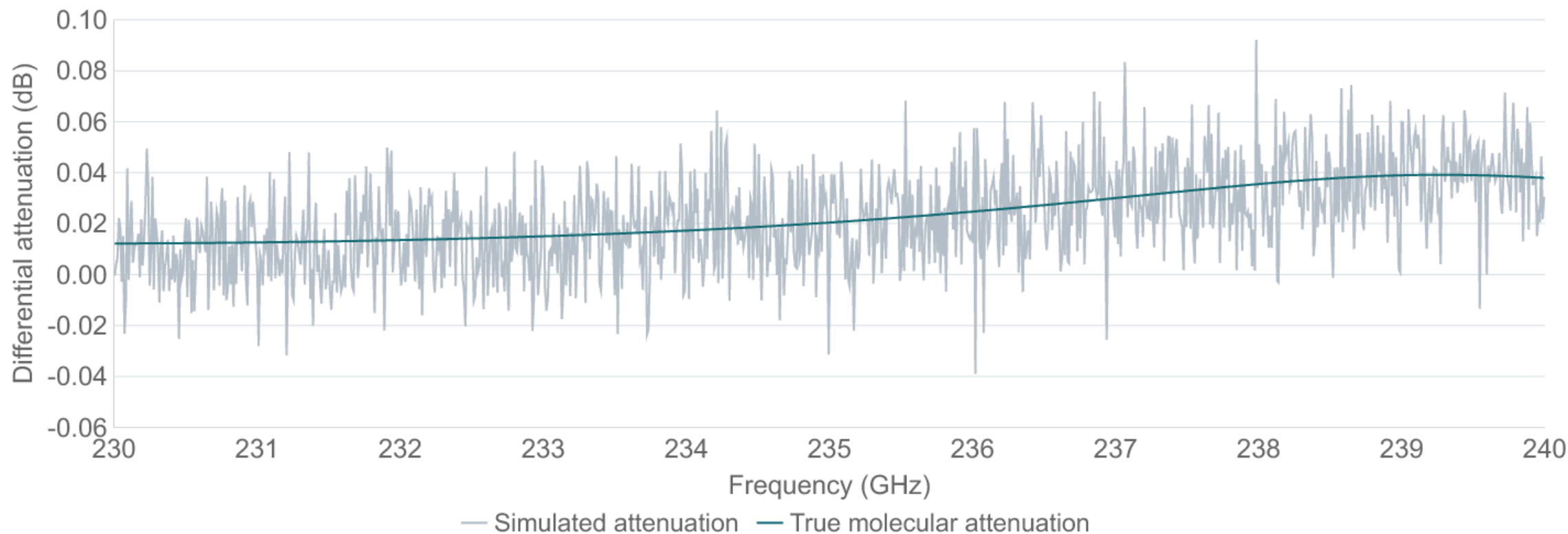
The 1% false alarm target applies to one prespecified compound and one decision.

A 95% detection concentration is higher than the decision threshold.

Solving $q - \tau = 1.64485 s_1(q)$ gives $q_{95} = 50.5814 \mu\text{g}/\text{m}^3$.

20 s total • ideal tracking • assumed, unmeasured 0.001 dB differential calibration residual

One simulated observation crosses the threshold



True $q = 50.5814$ · estimated $q = 39.4247 > 29.6174 \mu\text{g}/\text{m}^3 \rightarrow$ detection

20 s total • ideal tracking • assumed, unmeasured 0.001 dB differential calibration residual

Repeated simulations retain the failed low concentration case

10,000 independent reference/sample pairs for each concentration.

True q ($\mu\text{g}/\text{m}^3$)	Predicted response	Simulated response	95% interval
0	1.00%	1.13%	0.93–1.36%
1	1.23%	1.31%	1.10–1.55%
50.5814	95.00%	94.75%	94.29–95.18%

At 1 $\mu\text{g}/\text{m}^3$, the 1.31% simulated response is close to the false alarm rate.

20 s total • ideal tracking • assumed, unmeasured 0.001 dB differential calibration residual

The result does not establish background monitoring

The example predicts a 95% enhancement limit of $50.58 \mu\text{g}/\text{m}^3$ under its stated model.

An aircraft study reports a selected background of 145 parts per trillion by volume (pptv), about $0.252 \mu\text{g}/\text{m}^3$ at 288.15 K and 101,325 Pa.

The model limit is about 29.1 parts per billion by volume (ppbv) at those conditions, roughly 200 times that mixing ratio.

The useful result is an explicit detection calculation with a clear sensitivity limitation.

20 s total • ideal tracking • assumed, unmeasured 0.001 dB differential calibration residual

What is needed for a physical demonstration

Measure blank reference/sample stability and spectral covariance with the intended receiver.

Verify the available linear RF power and bandwidth, then test tracking during a moving pass.

Use independent concentration measurements and changing weather to assess false detections.

For 3D reconstruction, add distinct rays and evaluate matrix rank and spatial resolution.

LEO links remain a useful example for the complete chain, including its practical limits.

This tutorial models direct transmission. Molecular scattering and multistatic imaging need a separate feasibility study.

Appendix · Every entry of a five tone design matrix

Tone index k	$a_k [10^{-4} \text{ dB}/(\mu\text{g}/\text{m}^3)]$	1	u_k
0	2.408200	1	-0.500000
255	2.804245	1	-0.250733
511	4.032712	1	-0.000489
767	6.489083	1	0.249756
1023	7.500802	1	0.500000

$$\mathbf{y}_5 = \mathbf{A}_5 \begin{bmatrix} q & b_0 & b_1 \end{bmatrix}^T + \boldsymbol{\epsilon}_5, \quad \mathbf{A}_5 \in \mathbb{R}^{5 \times 3}$$

The table scales only the a column by 10^4 for readability. Use its original units in the calculation.

This subset is only an arithmetic illustration. The main result uses all 1,024 tones.

Appendix · The five tone covariance matrix

Diagonal entries contain thermal variance and calibration variance.

Off diagonal entries come from the assumed correlated calibration residual.

$C_5 / 10^{-4} \text{ dB}^2$	0	255	511	767	1023
0	2.951529	0.007796	0.006071	0.004728	0.003682
255	0.007796	2.947224	0.007788	0.006065	0.004724
511	0.006071	0.007788	2.947702	0.007788	0.006065
767	0.004728	0.006065	0.007788	2.952581	0.007788
1023	0.003682	0.004724	0.006065	0.007788	2.961572

C_5 is the corresponding submatrix of the null covariance $C(0)$.

Assumed, unmeasured 0.001 dB differential calibration residual · 20 s total acquisition

Appendix · Multiplication gives the five tone estimate

Tone index	Observed y_k (dB)	Recomputed H_5 weight
0	-0.000463815	4942.686
255	0.025099503	-4201.176
511	-0.031417175	-5681.920
767	0.018701897	4215.931
1023	0.030551022	724.479

$$\hat{q}_5 = \mathbf{H}_5 \mathbf{y}_5 = 171.7494, \quad s_{0,5} = 165.1163, \quad \tau_5 = 384.1179 \mu\text{g}/\text{m}^3$$

H_5 weights have units $(\mu\text{g}/\text{m}^3)/\text{dB}$. Both $\hat{q}_5$ and $s_{0,5}$ are in $\mu\text{g}/\text{m}^3$.

171.75 < 384.12: this five tone subset does not detect the same simulated sample.

20 s total · ideal tracking · assumed, unmeasured 0.001 dB calibration residual

Appendix · The spacing comparison

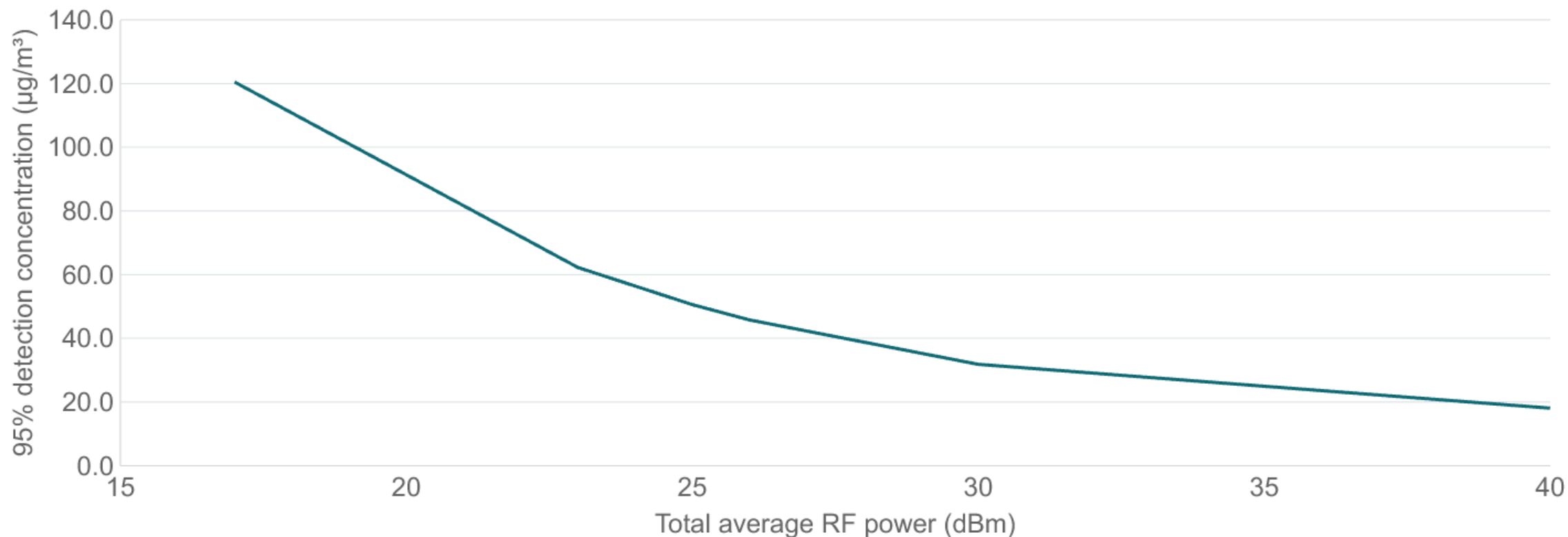
ICI means interference between tones. It is a separate screen, not included in the ideal q_{95} .

Tones	Δf (MHz)	CP time (%)	ICI at 100 kHz (%)	q_{95} ($\mu\text{g}/\text{m}^3$)
256	39.0625	28.09	0.00216	56.21
512	19.5312	16.34	0.00862	52.52
1024	9.7656	8.90	0.03449	50.58
2048	4.8828	4.66	0.13791	49.58

1,024 tones give the lowest ideal q_{95} among these cases with ICI below an illustrative 0.1% cap.

20 s total • ideal tracking • assumed, unmeasured 0.001 dB differential calibration residual

Appendix · More RF power helps, conditionally



25 dBm: $50.58 \mu\text{g}/\text{m}^3$ · 30 dBm: $31.83 \mu\text{g}/\text{m}^3$ · 40 dBm: $18.06 \mu\text{g}/\text{m}^3$

The 1 W and 10 W cases are unproven hardware scenarios.

20 s total · ideal tracking · assumed, unmeasured 0.001 dB differential calibration residual

Appendix · Calibration and background change the result

Change from the baseline	Predicted concentration scale
Zero added calibration residual	$q_{95} = 47.40 \mu\text{g}/\text{m}^3$
Assumed 0.001 dB residual	$q_{95} = 50.58 \mu\text{g}/\text{m}^3$
Assumed 0.01 dB residual	$q_{95} = 157.79 \mu\text{g}/\text{m}^3$
Fit an additional background amplitude	Local q_{95} approximation = $83.63 \mu\text{g}/\text{m}^3$

The error model and nuisance space determine what spectral information remains.

20 s total · ideal tracking · every calibration level is assumed and unmeasured

Appendix · The satellite moves during a measurement

Doppler is the frequency shift caused by motion between the transmitter and receiver.

Time after zenith	Elevation	Range (km)	Doppler at 235 GHz
0 s	90.000°	550.000	-0.0 kHz
0.5 s	89.605°	550.012	-37.8 kHz
5 s	86.053°	551.204	-377.0 kHz
25 s	70.886°	579.338	-1793.1 kHz

A coherent 10 s acquisition requires time dependent delay, phase, frequency and gain corrections.

The static sensitivity calculation assumes ideal tracking. It does not establish moving link performance.

Sources · Hardware and atmospheric context

Sen et al. (2023), Nature Electronics. Terrestrial 210–240 GHz link. DOI: [10.1038/s41928-022-00897-6](https://doi.org/10.1038/s41928-022-00897-6)

TeraLink (2026), preprint arXiv:2606.15410v1. Proposed LEO link parameters and qualification scope.

Cooper et al. (2025), IEEE Journal of Microwaves. 240 GHz frequency multiplier source. DOI: [10.1109/JMW.2025.3610360](https://doi.org/10.1109/JMW.2025.3610360)

EMeRGe study (2023), Atmospheric Chemistry and Physics 23, 1893. DOI: [10.5194/acp-23-1893-2023](https://doi.org/10.5194/acp-23-1893-2023)

Sources · Physics and reproducible calculations

HITRAN molecule 41, CH₃CN. Acquired line parameters and provenance are saved with the experiment.

ITU P.835-7 (2024): reference standard atmosphere. ITU P.676-13 (2022): gaseous attenuation and emission model.

Research snapshot b94f3fe: docs/54 and docs/55 contain the source audit and full derivation.

results/zenith_single_compound contains every tone, A, C, H, observations, sensitivity tables and hashes.

Reproduction entry point: `scripts/run_zenith_tutorial.py`